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DISMA

Anti-plane analysis of Francfort-Marigo model for quasi-static brittle fracture

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Master's Thesis in Engineering Mathematics

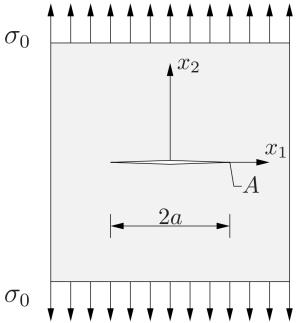
Historical setting

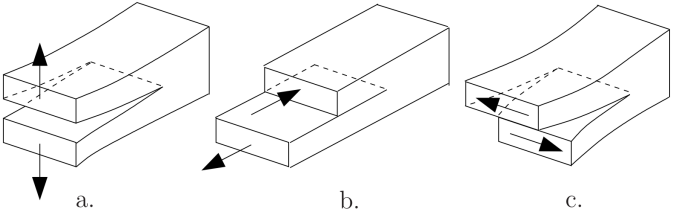
- Study and prediction of crack behavior is always central in solid mechanics.
- Seminal work by Griffith in 1921, who computed the critical imposed tension for a given crack to propagate:

$$\sigma_{0c} = \sqrt{\frac{2E'\gamma}{\pi a}}.$$

- Despite the model limitations, this led to Griffith's criterion, where G is the Energy Release Rate:

$$\begin{cases} G < G_c & \text{crack does not propagate,} \\ G = G_c & \text{crack propagates.} \end{cases}$$





- Irwin in 1958 introduced the modes of fracture, and deduced the singularity in displacement at the tip (κ is Stress Intensity Factor):

$$u_{ij} \propto \kappa\, r^{\frac{1}{2}} \implies \sigma_{ij} \propto \kappa\, r^{-\frac{1}{2}}.$$

- Rice and Cherepanov in 1968 introduced the path-independent J-integral, which led to the dissipative formulation of the quasi-static Griffith criteria:

$$\begin{cases} \dot{\ell} \geq 0, \\ (G_c - G) \geq 0, \\ (G_c - G)\,\dot{\ell} = 0. \end{cases}$$

Francfort-Marigo model

In 1998, Francfort and Marigo aimed at solving some problems which plagued Griffith's formulation, by incorporating an interplay between elastic and fracture energies and a minimization over admissible crack evolutions.

Interplay of energies

We define the total energy as a function of the loading g and the crack set K:

$$E(g,K) := \min_{v\in\mathcal{V}(g,K)} \left\{ \int_{\Omega\setminus K} W(x,\varepsilon(v)(x))\,\mathrm{d}x + \int_K k(x)\,\mathrm{d}\mathcal{H}^{N-1}(x) \right\};$$

$$\mathcal{V}(g,K) := \{v \in V(\Omega\setminus K) : v = g \text{ on } \partial_D\Omega\setminus K\}.$$

The elastic energy density $W(x,\xi)$ has to be a Carathéodory function, strictly convex and p-homogeneous in ξ, and such that $\alpha|\xi|^p \le W(x,\xi) \le \beta\big(|\xi|^p+1\big)$ with $\alpha,\beta>0$.

The energies do not depend explicitly on time, hence the evolution rule has to be added.

Quasi-static continuous evolution

A quasi-static evolution of a crack has to satisfy these conditions:

1. $K(t)$ is non-decreasing in t, and $K_0 \subseteq K(0)$;
2. $K(t)$ minimizes the energy among all possible crack such that
$$E(g(t), K(t)) \leq E(g(t), K) \qquad \forall K \supseteq \bigcup_{s<t} K(s);$$
3. The function $t \mapsto E(g(t), K(t))$ is absolutely continuous, and for a fixed loading $g(t)$, $K(t)$ is a stationary point for the function $s \mapsto E(g(t), K(s))$, that is
$$\frac{\mathrm{d}}{\mathrm{d}s} E(g(t), K(s))\Big|_{s=t} = 0 \qquad \text{for a.e. } t.$$

- **Strengths of the model**
  - Study of crack nucleation, crack path, crack failure.
  - Under certain assumptions, can obtain Griffith's criteria.
  - Can discern between continuous or discontinuous crack evolution.
  - Physically sound basis for the evolution.
- **Weaknesses of the model**
  - Can not handle imposed forces.
  - Requires complex numerical methods.
- **Possible extensions**
  - Dependence on the normal to the crack set, and to the jump in displacement, to incorporate more complex behaviors like Barenblatt's hypothesis.
  - Contact between the lips of the crack, with friction.
  - Ductile materials undergoing plastic deformation at the tip.

Two-dimensional anti-plane case and assumptions

For the proof of the existence of a continuous evolution, we will consider for simplicity a linearly elastic homogeneous isotropic material, undergoing a two-dimensional anti-plane shear. Moreover, we will assume the fracture toughness $k(x)$ to be a constant γ. This amounts to:

Simplified setting

$\Omega \subset \mathbb{R}^2$ is a bounded open set, such that $\partial\Omega = \partial_D\Omega \sqcup \partial_N\Omega$, K is a compact subset of Ω, $g \in H^1(\Omega)$, and μ is the shear modulus of the material, and γ the fracture toughness.

$$E(g,K) := \min_{v \in \mathcal{V}(g,K)} \left\{ \frac{\mu}{2} \int_{\Omega \setminus K} |\nabla v|^2 \, \mathrm{d}x + \gamma \, \mathcal{H}^1(K) \right\};$$

$$\mathcal{V}(g,K) := \left\{ v \in L^{1,2}(\Omega \setminus K) : v = g \text{ q.e. on } \partial_D\Omega \setminus K \right\}.$$

Objective: prove the existence of a compact-valued non-decreasing function $K\colon [0,1] \to \mathcal{K}^f_m(\overline{\Omega})$ satisfying the stability conditions of the continuous evolution, where $\mathcal{K}^f_m(\overline{\Omega})$ is the space of compact non-empty subsets of $\overline{\Omega}$ with finite $\mathcal{H}^1$-measure and at most m connected components.

Existence result

Let $g \in \mathrm{AC}\big([0,1]; H^1(\Omega)\big)$ and $K_0 \in \mathcal{K}^f_m(\overline{\Omega})$. Then there exists a continuous evolution $K\colon [0,1] \to \mathcal{K}^f_m(\overline{\Omega})$ such that:

1. $K_0 \subseteq K(s) \subseteq K(t)$ for $0 \le s < t \le 1$;
2. $E(g(0), K(0)) \le E(g(0), K) \quad \forall K \in \mathcal{K}^f_m(\overline{\Omega}), K \supseteq K_0$;
3. for $t \in (0,1]$,
   $E(g(t), K(t)) \le E(g(t), K) \quad \forall K \in \mathcal{K}^f_m(\overline{\Omega}), K \supseteq \bigcup_{s<t} K(s)$;
4. the function $t \mapsto E(g(t), K(t))$ is absolutely continuous on $[0,1]$;
5. $\dfrac{\mathrm{d}}{\mathrm{d}s} E(g(t), K(s))\Big|_{s=t} = 0$ for a.e. $t \in [0,1]$.

Preliminary concepts and results

Deny-Lions spaces

Only the integrability of the gradient is needed, hence we use the space of functions:

$$L^{1,p}(\Omega) := \left\{ v \in L^p_{\mathrm{loc}}(\Omega) : \nabla v \in L^p(\Omega; \mathbb{R}^N) \right\},$$

equipped with the norm $\|\nabla v\|_{L^p(\Omega;\mathbb{R}^N)}$.

Properties of Deny-Lions spaces

- This space is complete, hence $\left\{ \nabla v : v \in L^{1,p}(\Omega) \right\}$ is closed in $L^p(\Omega; \mathbb{R}^N)$.
- If Ω is bounded and with a Lipschitz boundary, then $L^{1,2}(\Omega) = H^1(\Omega)$.

Preliminary concepts and results

Hausdorff measure and Hausdorff metric

Hausdorff measure

Let $h(A)$ be a non-decreasing set function. Let $\mathcal{A}_\varepsilon$ be a covering of E with sets such that $h(A_i) < \varepsilon$. Then we define:

$$\Delta^h(E) := \sup_{\varepsilon \to 0} \Delta^h_\varepsilon(E) = \sup_{\varepsilon \to 0} \inf_{\mathcal{A}_\varepsilon} \sum_{A \in \mathcal{A}_\varepsilon} \alpha(h) h(A).$$

If $h(A) = \big(\mathrm{diam}(A)\big)^s$, we get the s-dimensional Hausdorff measure $\mathcal{H}^s(E)$.

Hausdorff distance

Let $\mathcal{K}(\overline{\Omega})$ be the collection of all compact non-empty subsets of $\overline{\Omega}$. When $X \supseteq \Omega$ is complete, then $\mathcal{K}(\overline{\Omega})$ is complete and compact with the distance:

$$\mathcal{D}(K_1, K_2) := \max\Big\{ \sup_{x \in K_1} d(x, K_2), \ \sup_{y \in K_2} d(y, K_1) \Big\}.$$

Preliminary concepts and results

Variational p-capacity

We define a set function as

$$\mathrm{Cap}_p(A;\Omega) := \inf_{u\in\mathcal{W}(A;\Omega)} \int_\Omega |\nabla u|^p \,\mathrm{d}x,$$
$$\mathcal{W}(A;\Omega) := \left\{ u \in W_0^{1,p}(\Omega) \cap C(\Omega) : A \subseteq \{u \geq 1\}^\circ \right\},$$

which can be proved to be an outer measure, useful in defining the following concepts:

- A property is valid *quasi-everywhere* if it is true outside of a set of zero capacity.
- A function is *quasi-continuous* if it is continuous outside of a set of zero capacity.
- A set E is said to be *invisible* for a given Sobolev space V if there exists an isometric isomorphism between $V(\Omega)$ and $V(\Omega \setminus E)$: this condition is equivalent to $\mathrm{Cap}_p(E) = 0$.

Method of proof

Proof obtained through a time discretization:

1. take a time-step $\delta > 0$, define

$$t_i^\delta := i\delta, \qquad g_i^\delta := g(t_i^\delta);$$

2. define K_i^δ inductively as

$$K_i^\delta := \operatorname*{argmin}_K \left\{ E\left(g_i^\delta, K\right) : K \in \mathcal{K}_m^f(\overline{\Omega}), K \supseteq K_{i-1}^\delta \right\}, \qquad K_{-1}^\delta := K_0$$

   and then u_i^δ as the solution to the minimum of $E(g_i^\delta, K_i^\delta)$;

3. define then the step functions on $[0, 1]$ by

$$g_\delta(t) := g_i^\delta, \qquad K_\delta(t) := K_i^\delta, \qquad u_\delta(t) := u_i^\delta, \qquad t \in \left[t_i^\delta, t_{i+1}^\delta\right).$$

Needed theorems

- A result guaranteeing that $E(g_n, K_n) \to E(g, K)$ if $g_n \to g$ in $H^1(\Omega)$ and $K_n \to K$ in the Hausdorff metric, as $n \to \infty$, or equivalently $\delta \to 0$, for the well-posedness of the discretized evolution.

- A sequence of compact-valued non-decreasing functions admits a limit K, such that $K_n(t) \to K(t)$ in the Hausdorff metric for every $t \in [0, 1]$, for ensuring the existence of continuous non-decreasing evolution in the limit.

- Convergence of $K_\delta(t) \to K(t)$ in $\mathcal{K}^\lambda_m(\overline{\Omega})$ in the Hausdorff metric, and $\nabla u_\delta(t) \to \nabla u(t)$ in $L^2(\Omega; \mathbb{R}^2)$, for every $t \in [0, 1]$ and as $\delta \to 0$, to prove the existence of the continuous evolution given the properties of the discretized evolution.

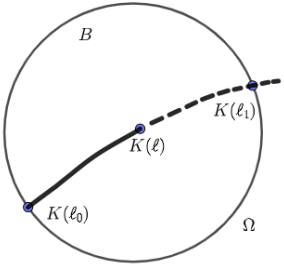
Behaviour of displacements near the crack tip

Objective: deduce Irwin's singularity at the crack tip, and study the Stress Intensity Factor.

If we parameterize the crack with its length $K = K(\ell)$, we have

$$u - \frac{\kappa}{\mu} r^{\frac{1}{2}} \cos\frac{\theta}{2} \in H^2(\Omega \setminus K(\ell)) \cap H^{1,\infty}(\Omega \setminus K(\ell)),$$

where κ is proportional to the Stress Intensity Factor.



This requires a study of the properties of solutions to Dirichlet problems in polygonal boundaries. Conclusions are obtained through Fredholm's alternative theory, by studying the Laplacian as an operator from the space of strong solutions to $L^2(\Omega)$.

When the crack is allowed to propagate, we get

$$\frac{\mathrm{d}}{\mathrm{d}\ell} E(g, K(\ell)) = \underbrace{\gamma}_{= G_c} - \underbrace{\frac{\pi}{2} \frac{\kappa(u(\ell), \ell)^2}{2\mu}}_{= G},$$

where $u(\ell)$ is a solution to the minimum of $E(g, K(\ell))$.

Griffith's criteria

Under the assumptions of the existence theorem, we have:

$$\begin{aligned} \dot{\ell}(t) &\geq 0 && \text{for a.e. } t \in (0,1), \\ \gamma - \frac{\pi}{2} \frac{\kappa(u(\ell(t)), \ell(t))^2}{2\mu} &\geq 0 && \text{for every } t \in (0,1), \\ \left\{ \gamma - \frac{\pi}{2} \frac{\kappa(u(\ell(t)), \ell(t))^2}{2\mu} \right\} \dot{\ell}(t) &= 0 && \text{for a.e. } t \in (0,1). \end{aligned}$$

Thank you for your attention

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